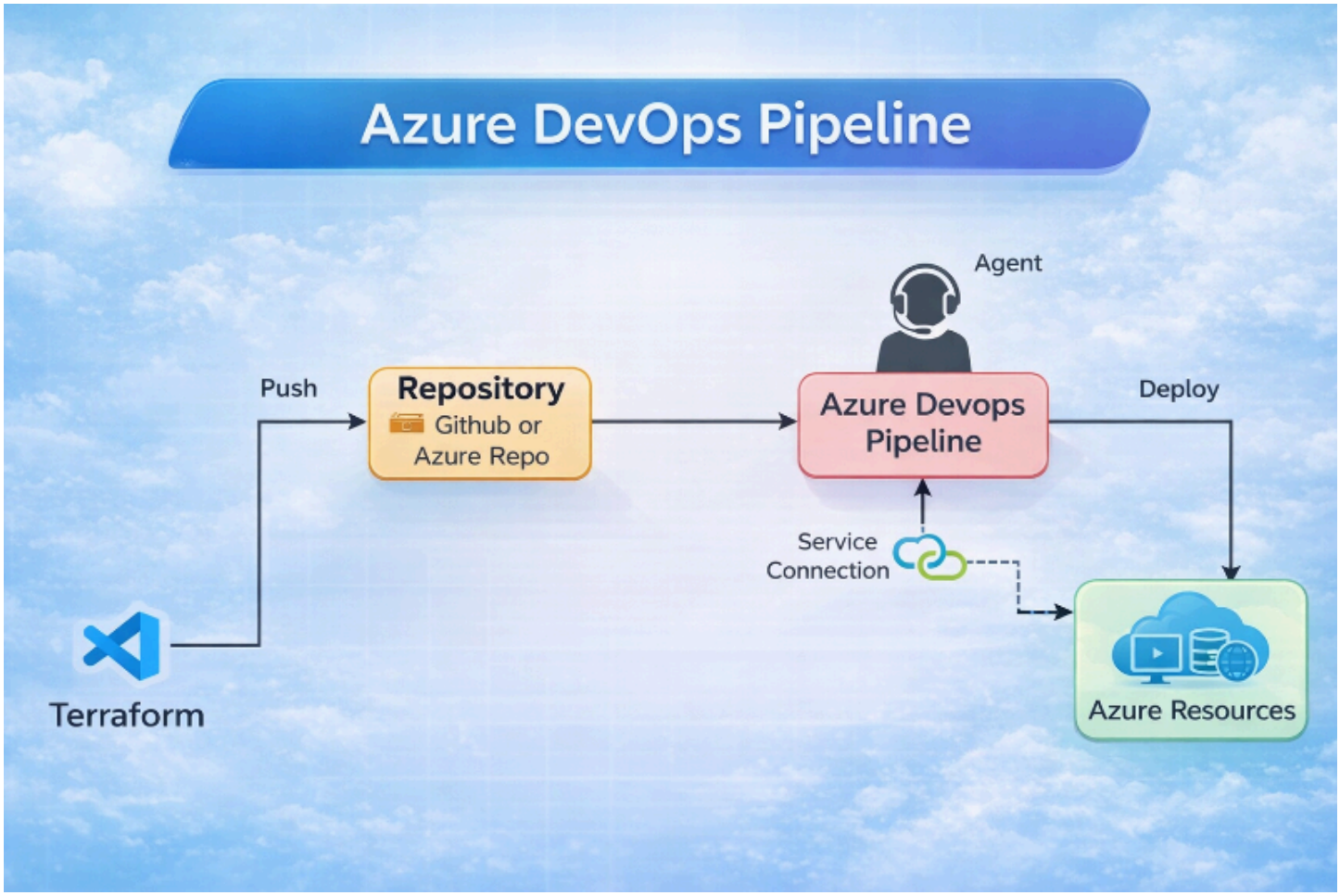
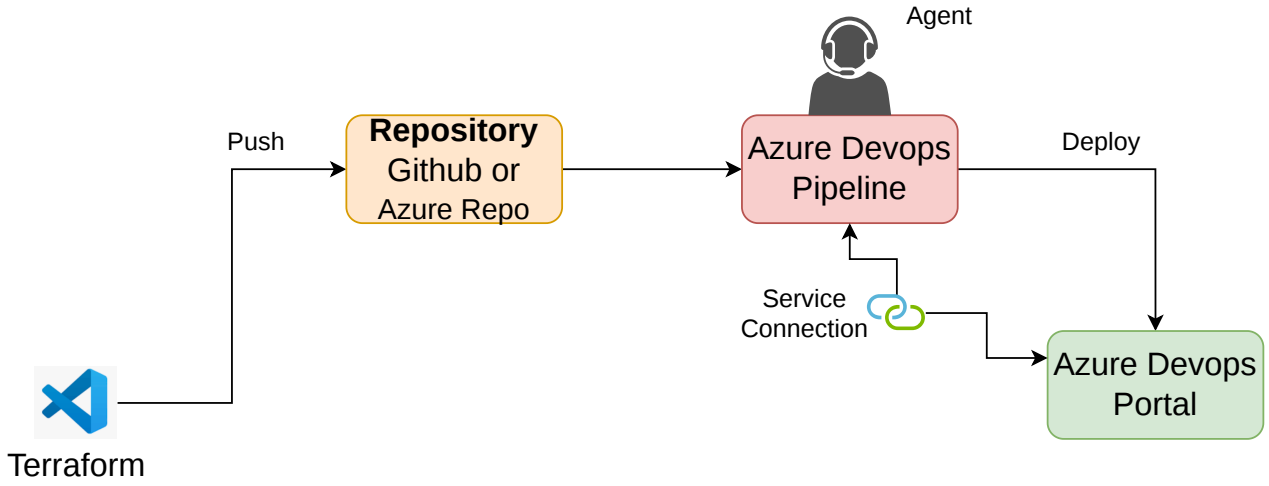


Azure Devops Pipeline



Terraform Code

```
terraform {
  backend "azurerm" {}
  required_providers {
    azurerm = {
      source  = "hashicorp/azurerm"
      version = "4.66.0"
    }

  }
}

provider "azurerm" {
  features {}
}


resource "azurerm_resource_group" "rg" {
  name     = "sanjay_rg"
  location = "eastus"
}
```

Github

Login-https://github.com/
Create Repo

In VS Code
git init
git add .
git commit -m "Initial terraform code"
git remote add origin <repo>
git branch -M main
git push -u origin main

Next Time (Already repo exists)
git add .
git commit -m "updated terraform code"
git push

Next Time (Already repo exists- Feature Branch ke sath)

Azure Devops (ADO)

Login: https://dev.azure.com/
Create Organization
Create Project
Create Agent Pool
Create Agent
Create Service Connection
Create Storage Account

Agent Pool & Agent

Create Agent Pool---Project Setting----Agent Pools----Add Pool---Self Hosted---Assign name---- Give Permission--- Create

Create Agent----Under Agent Pools----New Agent----Download agent----configure agent----run agent

Service Connection

Project Setting
New Service Connection----ARM
Identity type-----Manual

Azure Portal App Registration
Name----Anything then create.
Client credentials-----Add a certificate or secret
Federated credentials----Add credentials

Credential----- --Work Load Identity Federation **Service Connection Name----anything** **Environment-----Azure Cloud** **Directory (tenant) ID----Azure Portal under Microsoft Intra ID** **Next** **Subscription ID-----Azure Portal under subscription** **Subscription name---Azure Portal under subscription** **Application (client) ID----Azure Portal App Registration** **Grant Permission** **Verify and Save.**	Federated credential scenario----Other Issuer Issuer and Value----from ADO (app registration details) name----anything Add. **Assign the contributor role to app registration** **Subscription----IAM----Add role assignment** **Privileged Administrator roles----contributor----select member-----Give the App Reg. Name----Review & Assign.**

Create Storage Account

Azure Portal-----Storage Account----Create
Under storage Account------Create Container
Insider Container----Add role assignment----Job Function role----Storage blob data contributor----Select App Registration.

Basic Infra YAML Pipeline

```yaml
trigger: none

pool: B17_AgentPool

stages:
  - stage:
    jobs:
      - job:
        steps:
          - task: TerraformInstaller@1
            inputs:
              terraformVersion: 'latest'
          - task: TerraformTask@5
            inputs:
              provider: 'azurerm'
              command: 'init'
              backendServiceArm: 'xyz'
              backendAzureRmStorageAccountName: 'sanjaystg1'
              backendAzureRmContainerName: 'xyz'
              backendAzureRmKey: 'terraform.tfstate'
          - task: TerraformTask@5
            inputs:
              provider: 'azurerm'
              command: 'plan'
              environmentServiceNameAzureRM: 'xyz'
          - task: TerraformTask@5
            inputs:
              provider: 'azurerm'
              command: 'apply'
              environmentServiceNameAzureRM: 'xyz'
```